

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms.

Win condition: Best lift with correct setup and explanation.

THE SCIENCE YOU ARE USING

Mechanical advantage. A single fixed pulley just changes the direction of your pull. Add movable pulleys and you trade distance for force: you pull more rope but with less effort. More supporting rope segments means more mechanical advantage.

Force and direction. Pulleys let you pull down to lift up, and let a small force raise a big load. You build pulley systems, compare the effort needed, and explain why the setup helps.

HOW TO BUILD AND RUN IT

- 1 Lift with no pulley.** Measure the effort to lift the load by hand as a baseline.
- 2 Add a fixed pulley.** Reroute over one fixed pulley. Note that direction changes but effort does not.
- 3 Add a movable pulley.** Build a system with a movable pulley and feel the effort drop.
- 4 Measure the advantage.** Use a spring scale to compare effort across setups.
- 5 Explain the tradeoff.** Show how more rope segments cut the force you need.

Safety: Keep feet clear of loads. Allergy: None.

MATERIALS

Classroom pulleys	per team
String	shared
Weights	shared
Spring scales	if available

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Performance	35
Setup accuracy	25
Explanation	20
Teamwork	20
Total	100